

GARRETT COMES

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EDUCATION

Georgia Institute of Technology

Jul 2022 – Aug 2026

B.S. Aerospace Engineering, Concentration in Propulsion

Atlanta, GA

- Coursework: Jet & Rocket Propulsion, Computational Fluid Dynamics, Aerothermodynamics, Finite Element Analysis, Control System Design, Aircraft Flight Dynamics, Aerodynamics.

EXPERIENCE

Lockheed Martin Space — Propulsion Engineering Intern, Civil & Commercial Space

Summers 2022 – 2025

Flight programs incl. Dragonfly and Mars Sample Return

Waterton Campus, Denver, CO

- Designed a piston-tank propellant gauging system in which hermetically sealed internal magnets are read by radiation-hardened Hall-effect sensors, owning magnetic circuit sizing, sensor selection, and the integration concept, and ran magnetic simulations with optimization methods to size the design against any given set of requirements.
- Ran propellant slosh and fluid-structure-interaction simulations of spacecraft tanks in Ansys, plus pure slosh CFD in FLOW-3D, and supported design of a slosh characterization test rig used to correlate model predictions against test data.
- Contributed to a green-propellant trade study evaluating hydrazine-replacement monopropellants across performance, materials compatibility, and handling and safety criteria.
- Designed and analyzed propulsion hardware and fluid-system components, including valves and test support equipment, with Creo CAD models and drawings, and supported test campaigns, requirements verification, and anomaly investigation.

PROJECTS & RESEARCH

SCRAP — Orbital Debris Detumbling & Capture, Senior Design | *Python, FEniCSx, Rust, Gmsh*

2025 – 2026

- Co-authoring an AIAA SciTech 2027 extended abstract on a servicing spacecraft that detumbles and captures orbital debris with a robot-arm-mounted YBCO high-temperature-superconducting coil.
- Developed a coupled electromagnetic-thermal quench simulation of the HTS pancake coil, using an H-formulation with E-J power-law resistivity, per-turn tape meshing, and radiation/cold-finger BCs, to establish safe operating current margins.
- Built [spacedb](#), a Rust space-object database and GUI with collision and decay analysis, target value scoring, and multi-target servicing route optimization over the full public catalog.
- Trained DDPG reinforcement-learning agents for eddy-current detumbling of uncooperative targets.

Valurile — GPU Lattice-Boltzmann CFD Solver | *Rust, CUDA*

2024 – present

- Wrote a multi-crate LBM solver implementing Esoteric Pull in-place streaming, roughly halving memory traffic per timestep, plus static mesh refinement, Smagorinsky-Lilly LES, and SRT/TRT/MRT collision operators.
- Built an interactive egui/GPU frontend with airfoil and cylinder obstacles, physical-to-lattice nondimensionalization, live velocity/density/pressure/vorticity fields, and drag/lift coefficient histories.

Thermodynamic Sampling Research | *Python, JAX, PyTorch*

2026

- Authored a preprint measuring when thermodynamic sampling hardware (Extropic's TSU) beats GPUs on the energy cost of protein and drug design, showing about $10^6\times$ less sampling energy for equal library coverage across 59 randomized Potts landscapes, then honestly bounding the end-to-end ratio by shared digital readout and matching.
- Earlier, hybridized a pretrained masked-diffusion language model with THRML block-Gibbs joint decoding, beating independent argmax by 7–15 pp on held-out data, with bit-for-bit reproduction, 9/9 regression tests, and review gates on every headline claim.

fem_rust_ae — Finite-Element Solver in Rust, live in the browser | *Rust, WASM*

2025

- Implemented an FEM structural solver from scratch with mesh convergence studies; compiled to WebAssembly for an interactive in-browser demo.

SKILLS

Languages: Python, Rust, C++, MATLAB, CUDA/Triton, LaTeX

Simulation & CAE: Ansys Fluent and Mechanical, FEniCSx, Gmsh, SolidWorks, Creo, Fusion 360, Inventor

Scientific computing: JAX, PyTorch, NumPy/SciPy, wgpu, MPI/PETSc, engineering optimization

Tooling: Linux/NixOS, Git, agentic AI development with Claude Code, MCP servers, and custom agent frameworks

LEADERSHIP & CERTIFICATIONS

Yellow Jacket Space Program, 2023 – 2024, pressurant isolation valve design for a collegiate liquid-propellant space-shot vehicle.

Certifications: SolidWorks CSWA · Autodesk Inventor · Autodesk Fusion 360 · Wolfram Language Level 1